

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : SCALA FOR DATA SCIENCE

MODULE 1

PRACTICAL NO - 11, 12, 13, 14, 15, 16

Aim :

11. Write a program to tokenize and count the frequency of words in a text file.
12. Implement one-hot encoding for a categorical column in a dataset.
13. Create a scatter plot of random data using Breeze-viz. Label the axes and customize the color of points.
14. Generate a histogram of a dataset using Breeze-viz. Experiment with different bin sizes.
15. Plot a line graph for a dataset showing a trend over time.
16. Combine two plots (e.g., scatter and line plot) in a single visualization using Breeze-viz.

Practical 11

CODE :

```
import scala.io.Source
object omwala_practical11 {
  def main(args: Array[String]): Unit = {
    val filename = "om.txt"

    val lines = Source.fromFile(filename).getLines().toList

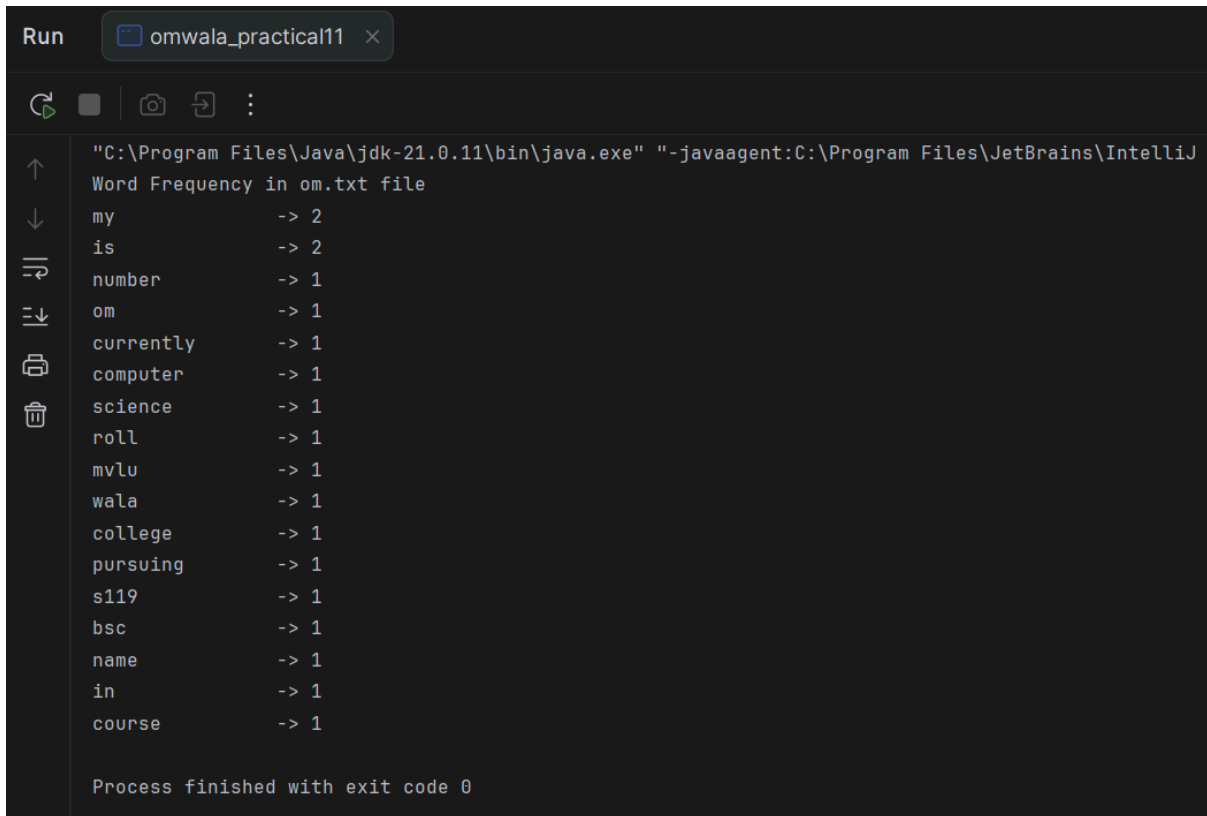
    val words = lines
      .flatMap(_.toLowerCase.split("\\W+"))
      .filter(_.nonEmpty)

    val wordcount = words.groupBy(identity).view.mapValues(_.size).toMap

    println("Word Frequency in om.txt file")
    wordcount.toSeq.sortBy(-_._2).foreach { case (word, count) =>
      println(f"$word%-15s -> $count")
    }
  }
}
```

OUTPUT :

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```
Run omwala_practical11 x
"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ
Word Frequency in om.txt file
my          -> 2
is          -> 2
number      -> 1
om          -> 1
currently   -> 1
computer    -> 1
science     -> 1
roll        -> 1
mvlv        -> 1
wala        -> 1
college     -> 1
pursuing    -> 1
s119        -> 1
bsc         -> 1
name        -> 1
in          -> 1
course      -> 1

Process finished with exit code 0
```

Practical 12

CODE :

```
import com.github.tototoshi.csv._
import java.io.File
object omwala_practical12 {
  def main(args: Array[String]): Unit = {
    val reader = CSVReader.open(new File("Tariff.csv"))
    val data = reader.allWithHeaders()
    reader.close()

    val categories = data.map(_("Product Type")).distinct.sorted
    val newData = data.map { row =>
      val product_type = row("Product Type")
      val oneHot = categories.map(cat => cat -> (if (cat == product_type) "1"
        else "0")).toMap
      (row - "Product Type") ++ oneHot
    }

    val headers = newData.head.keys.toList
    println(headers.mkString(", "))

    newData.foreach { row =>
      println(headers.map(row).mkString(", "))
    }
  }
}
```

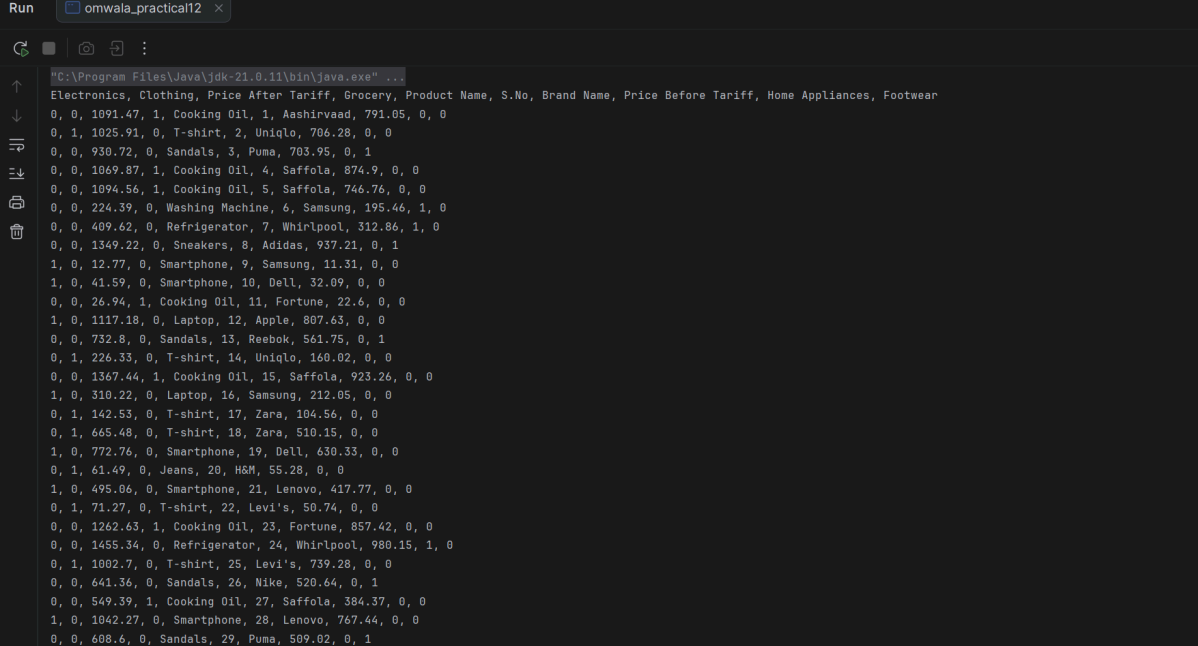
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```
val writer = CSVWriter.open(new File("Tariff_encoded.csv"))
writer.writeRow(headers)
newData.foreach(row => writer.writeRow(headers.map(row)))
writer.close()
println("One-hot encoded file written to Tariff_encoded.csv")
}
```

OUTPUT :



Practical 13

CODE :

```
import breeze.linalg._
import breeze.plot._
import com.github.tototoshi.csv._
import java.io.File

object omwala_practical13 {
  def main(args: Array[String]): Unit = {
    val reader = CSVReader.open(new File("Tariff.csv"))
    val data = reader.allWithHeaders()

    reader.close()

    val Samsung = data.filter(_("Brand Name") == "Samsung")
    val Levis = data.filter(_("Brand Name") == "Levi's")
    val Zara = data.filter(_("Brand Name") == "Zara")
    def extractXY(rows: List[Map[String, String]]) = {
      val x = DenseVector(rows.map(_("Price Before Tariff")).toDouble).toArray
      val y = DenseVector(rows.map(_("Price After Tariff")).toDouble).toArray
    }
  }
}
```

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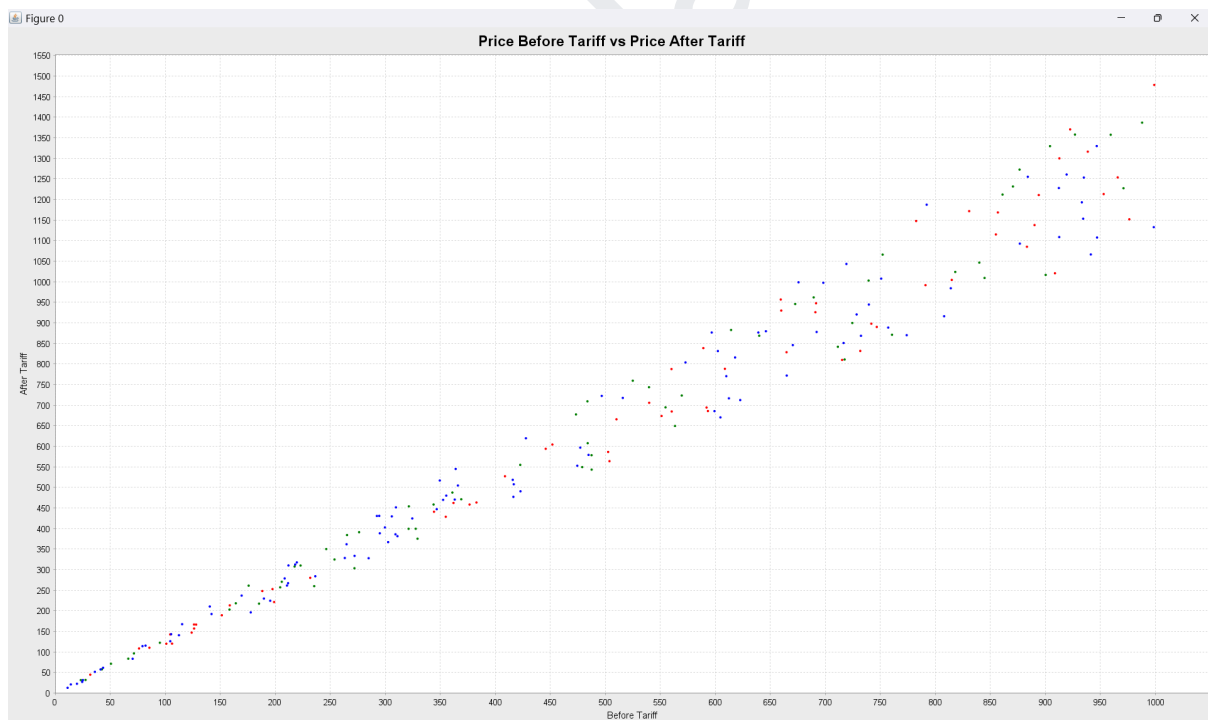
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```
(x, y)
}
val (xSamsung, ySamsung) = extractXY(Samsung)
val (xLevis, yLevis) = extractXY(Levis)
val (xZara, yZara) = extractXY(Zara)

val fig = Figure()
val plt = fig.subplot(0)
plt.title = "Price Before Tariff vs Price After Tariff"
plt.xlabel = "Before Tariff"
plt.ylabel = "After Tariff"
plt += plot(xSamsung, ySamsung, '.', name = "Samsung", colorcode = "blue")
plt += plot(xLevis, yLevis, '.', name = "Levis", colorcode =
    "green")
plt += plot(xZara, yZara, '.', name = "Zara", colorcode =
    "red")
fig.refresh()
}
}
```

OUTPUT :



Practical 14

CODE :

```
import breeze.linalg._
import breeze.plot._
import com.github.tototoshi.csv._
```

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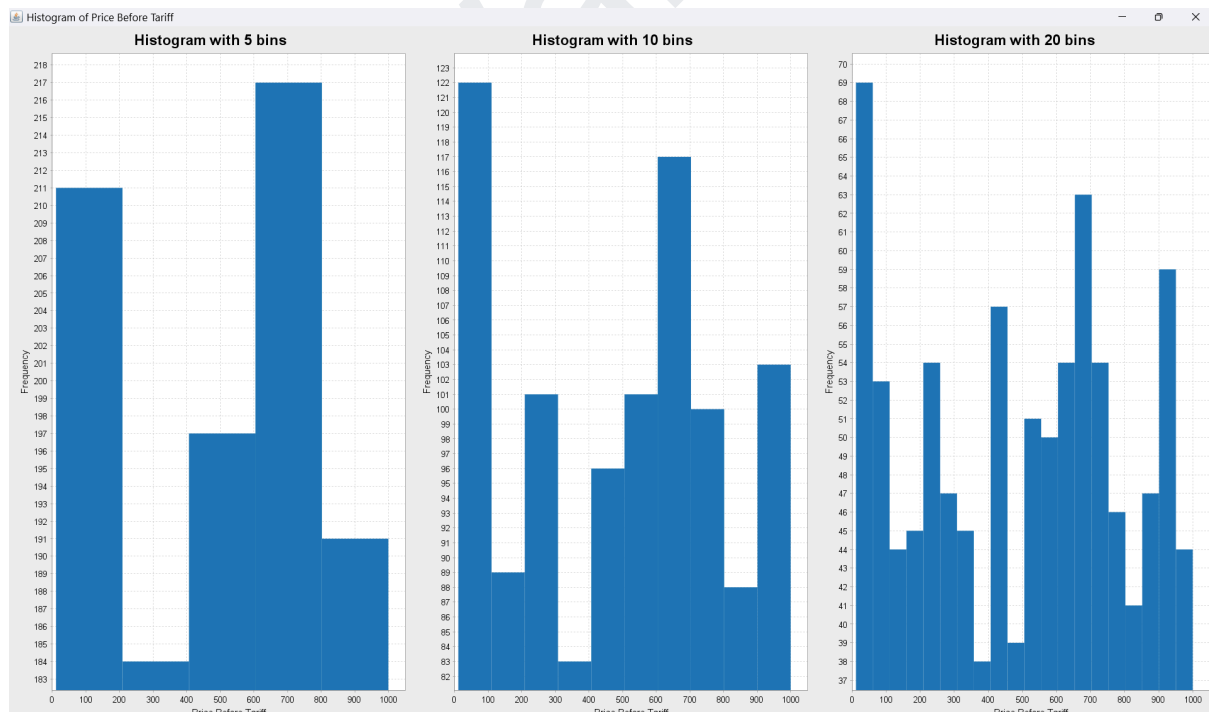
```
import java.io.File
object omwala_practical14 {
  def main(args: Array[String]): Unit = {
    val reader = CSVReader.open(new File("Tariff.csv"))
    val data = reader.allWithHeaders()
    reader.close()

    val BeforeTariff =

      DenseVector(data.map(_("Price Before Tariff").toDouble).toArray)

    val fig = Figure("Histogram of Price Before Tariff")
    val binSizes = List(5, 10, 20)
    for ((bins, idx) <- binSizes.zipWithIndex) {
      val plt = fig.subplot(1, binSizes.length, idx)
      plt += hist(BeforeTariff, bins)
      plt.title = s"Histogram with $bins bins"
      plt.xlabel = "Price Before Tariff"
      plt.ylabel = "Frequency"
    }
    fig.refresh()
  }
}
```

OUTPUT :



Practical 15

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CODE :

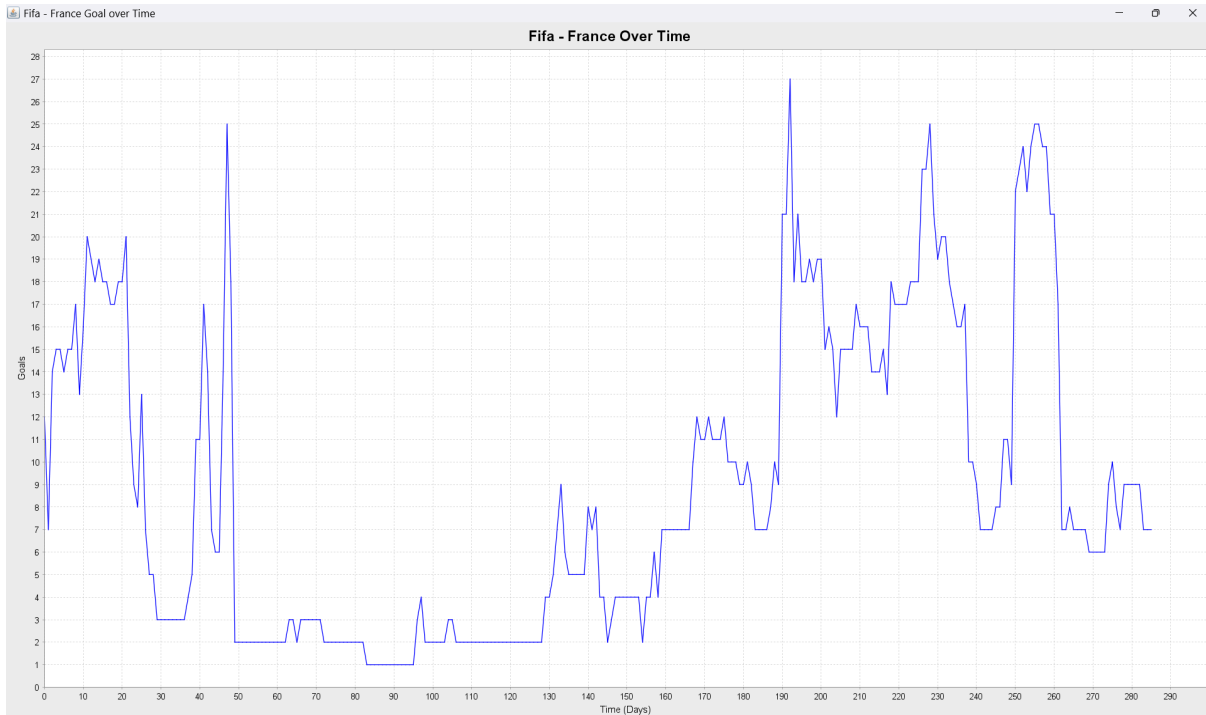
```
import breeze.linalg._
import breeze.plot._
import com.github.tototoshi.csv._
import java.io.File
import java.time.LocalDate
import java.time.format.DateTimeFormatter

object omwala_practical15 {
  def main(args: Array[String]): Unit = {
    val reader = CSVReader.open(new File("fifa.csv"))
    val data = reader.allWithHeaders()
    reader.close()

    val dateFormatter = DateTimeFormatter.ofPattern("yyyy-MM-dd")
    val parsedData = data.flatMap { row =>
      try {
        val date = LocalDate.parse(row("Date"), dateFormatter)
        val goal = row("FRA").toDouble
        Some((date, goal))
      } catch {
        case _: Throwable => None
      }
    }.sortBy(_._1)
    val x = DenseVector((0 until
      parsedData.length).map(_._2.toDouble).toArray)
    val y = DenseVector(parsedData.map(_._2).toArray)
    val fig = Figure("Fifa - France Goal over Time")
    val plt = fig.subplot(0)
    plt += plot(x, y, name = "Goal", colorcode = "blue")
    plt.xlabel = "Time (Days)"
    plt.ylabel = "Goals"
    plt.title = "Fifa - France Over Time"
    fig.refresh()
  }
}
```

OUTPUT :

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Practical 16

CODE :

```
import breeze.linalg._
import breeze.plot._
import com.github.tototoshi.csv._
import java.io.File
import java.time.LocalDate
import java.time.format.DateTimeFormatter
object omwala_practical16 {
  def main(args: Array[String]): Unit = {
    val reader = CSVReader.open(new File("fifa.csv"))
    val data = reader.allWithHeaders()
    reader.close()

    val dateFormat = DateTimeFormatter.ofPattern("yyyy-MM-dd")
    val parsedData = data.flatMap { row =>
      try {
        val date = LocalDate.parse(row("Date"), dateFormat)
        val goals = row("FRA").toDouble
        Some((date, goals))
      } catch {
        case _: Throwable => None
      }
    }

    }.sortBy(_._1)
```

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```
val x = DenseVector((0 until parsedData.length).map(_._1.toDouble).toArray)
val y = DenseVector(parsedData.map(_._2).toArray)
val fig = Figure("Fifa - Line + Scatter Plot")
val plt = fig.subplot(0)

plt += plot(x, y, name = "France Goal Line", colorcode = "blue")

plt += plot(x, y, '.', name = "France Goal Points", colorcode = "red")
plt.xlabel = "Time (Days)"
plt.ylabel = "Goals"
plt.title = "Fifa France Goals - Line + Scatter"
fig.refresh()
}
```

OUTPUT :

